

KUNAL RAI

Full-Stack Developer | Real-Time Systems | AI Applications

New Delhi, India

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Education

Bennett University

Bachelor of Technology in Computer Science **GPA: 8.08/10**

Sep 2023 – Sep 2027

Greater Noida, Uttar Pradesh

Relevant Coursework

- Data Structures
- Database Management
- FullStack
- Computer Networks
- Analysis of Algorithms
- Artificial Intelligence
- System Design
- Software Engineering

Projects

IdeaRoom – Real-Time Collaborative Whiteboard | *Next.js, Node.js, Socket.IO, MongoDB, Redis, RabbitMQ, Docker*

- Engineered a **real-time collaborative whiteboard platform** using **Socket.IO**, enabling seamless multi-user drawing and live canvas synchronization with **sub-100ms update latency**.
- Architected a **4-service microservices backend** (API Gateway, Auth, Collaboration Engine, Snapshot Worker), enabling independent scaling and improving service fault isolation.
- Improved data retrieval performance by **40%** by introducing **Redis caching** over MongoDB while offloading snapshot writes through **RabbitMQ event queues**.
- Containerized backend services using **Docker** and deployed production infrastructure on **Vercel and Railway**, enabling reliable continuous deployments.
- **GitHub:** github.com/VampKunal/IdeaRoom **Live:** idea-room-ashy.vercel.app

EKLAVYA – AI-Powered Voice Tutor | *Next.js, TypeScript, OpenAI API, Web Speech API*

- Developed an **AI-powered tutoring platform** integrating voice interaction, adaptive quizzes, and personalized lessons to create a hands-free learning experience.
- Integrated **speech recognition and text-to-speech pipelines** enabling students to interact with learning modules through voice commands.
- Designed an **adaptive difficulty system** that dynamically adjusts quiz complexity based on rolling student performance metrics.
- Leveraged **AI APIs** to automatically generate personalized study material and practice questions, reducing manual content creation.
- **GitHub:** github.com/VampKunal/EKLAVYA

VIBLY – Real-Time Chat & Video Calling Platform | *React, Node.js, MongoDB, Stream APIs, Zustand, TailwindCSS*

- Developed a **real-time messaging and video communication platform** using **Stream APIs**, enabling low-latency chat and HD video calls without managing WebRTC infrastructure.
- Implemented **friend-request workflows, online presence indicators, and customizable user themes** to enhance social interaction features.
- Optimized client-side state management using **Zustand** and reduced redundant API requests with **React Query caching**.
- **GitHub:** github.com/VampKunal/VIBLY

Technical Skills

Languages: C++, Python, JavaScript, TypeScript
Frontend: HTML5, CSS3, React, Next.js, Tailwind CSS
Backend: Node.js, Express.js
Databases: MongoDB, PostgreSQL, Redis
Tools: Git, GitHub, Docker, RabbitMQ

Problem Solving & Competitive Programming

- Solved **400+ Data Structures and Algorithms problems** on **LeetCode** covering Dynamic Programming, Graphs, Greedy Algorithms, and Binary Search.
- Ranked in the **Top 11% (4509 / 40,000 participants)** in a global competitive programming contest.